APSE Project: Crude Oil Pooling Problem

Jorge Urruela, CMU ID: jurruela, Project Option 1

Literature Review of Related Works

The problem statement (Castro, Crude Oil Pooling Problem, 2016) provides a general outline for crude oil pooling problems. It also provides a set of different GAMS files, each for different variables of the same problem, where parameters and units are different.

In this problem, a continuous time grid is used, which has the advantage that it requires less time slots than its discrete based counterparts (Castro, Crude Oil Pooling Problem Overview, 2016). However, this makes the problem horizon based, which is not required by traditional discrete grid. So an estimation of when the problem can and will end is required to begin.

Continuous time grids is a new formulation, which can improve the optimal setup to increase profit by even 67% in some cases for multipurpose plant batch assignment (Li, Rakovitis, Zheng, PanJie, & Kopanos, 2022) when compared to existing optimization formulations that include discrete time grids. There are other cases in which discrete time grid formulation is not optimal, and it can reduce computational effort by one order of magnitude.

For instances of crude oil pooling, it has been shown that this modeling has potential by reducing the number of slots. In a similar problem to the one in this study, it has been shown that it can improve the optimization value by 5%, while taking less than 1% of the time required when compared to a discrete time series solution with 16 slots (Mouret, Grossman, & Pestiaux, 2010).

Previously, to compensate for not finding the best solution with a discrete time problem, more and shorter time intervals were set. In one case (Floudas & Lin, 2004) increasing the number of slots from 8 to 32 enabled improving the objective by almost double, from 620 to 1195. However, the scaling for the computational power was not good, requiring solving for 591 variables instead of 38, and requiring now more than a day to find a solution, when previously it solved at fractions of a second. The same problem in a continuous time scale obtained an optimal value of 1498 in 0.28 seconds, requiring solving only for 40 binary variables.

Problem Statement

The problem (Castro, Crude Oil Pooling Problem, 2016) attempts to optimize a network in which crude oil from maritime vessels must be stored and charged before entering the distillation columns. The idea is that a maritime vessel carrying oil discharges all of its load into a storage tank. There are multiple vessels and multiple storage tanks, but only one vessel can discharge to its assigned storage tank. These storage tanks later discharge to the charging tanks, and in some variations, not all storage tanks are connected to all charging tanks. These discharge tanks contain stock to feed the distillation units, and again, in some variations of the problem not all discharge tanks are connected to every distillation unit. Also, some crude types cannot be processed in certain distillation units, hence, some crudes are not available for storage in their respective charging tanks.

A few logistic restrictions are assumed, to simplify the problem but based on reasonable operating conditions:

- Maritime vessels contain only one type of crude, and they are unloaded sequentially into one storage tank.
- Tanks containing different types of crude cannot be charged and discharged simultaneously. This would increase the problem complexity significantly, as concentrations are constantly changing without regulation.
- Charging tanks can be connected to multiple distillation units, but they can only unload to one distillation unit at a time
- Distillation units are being charged at all times, but their charging tank source varies over time, and each distillation unit 'selects' from one of exactly two discharge tanks
- The sequence is always maritime vessel->storage tank->discharge tank->distillation unit. A vessel cannot unload directly to a discharge tank or distillation unit, and a storage tank cannot provide directly to a distillation unit. This sequence must be followed

Also, different variations of the problem suggest either minimizing operation cost or maximizing profit, both logic objectives, but maximizing profit adds more complexity as demand does not have to be necessarily met if it does not make sense financially, while minimizing cost is tied to a certain demand of product as a constraint.

Motivation and Main Features of the Model

The motivation of the project is a project that maximizes profitability. And, as mentioned before, this can either be done by minimizing the operational cost or maximizing the profit.

These two achieve the same goal, but by simply minimizing the cost it is assumed that the product will be sold and that a certain demand must be met, while incorporating income does not necessarily assume that demands must be met, only as long as it makes financial sense. Both are very valid, as contracts and sales are negotiated in different ways according to the needs of the client, which opens the possibility of either alternative.

Each of these alternatives has its own complexities and features. First, maximization of profit features the sum of the individual gross margin of each distilled crude, which takes demand into consideration as well as price of individual types of oil, but disregards operating costs connected to the time that oil remains in a unit. It is also constrained to a maximum amount of distillation unit source changes.

On the other hand, minimization of operational cost does consider the cost of residence times in each unit, and penalizes with cost the distillation units changing of charging source, but is set to meet demands as a constraint, even to the point where it would not make financial sense because of the cost surpassing the price.

Some features are listed as follows:

- Storage and discharge tanks have a known initial inventory, a maximum capacity and a minimum capacity constraint.
- Flowrates between units are subject also to bounds (which make sense as pipes and pumps are also subject to capacities).
- Concentrations of crude types are maintained linear by computing the amounts of each crude type in each operation
- Transfer to columns are strictly positive
- Demands for crudes are bounded and set in the charging tanks

The model provides the option to be solved by 2 different methods: Baron or GloMIQO, and in here, the focus is toward minimizing operational cost, as it takes more into account and it is valid to assume that selling price is always higher than cost and demand always suffices.

Model Formulation

Sets (Seen later as subindices)

t: time sets, aliases tl
cr: crudes
mv: marine vessels
st: storage tanks

ct: charging tanks
cd: distillation units
pr: crude properties
tk: storage and charging tanks
u: units (tanks, vessels, distillation units), aliases ul, ull
btk_{tk}: blending tanks (tanks containing multiple crude types)
stk_{tk}: storage tanks
ctk_{tk}: charging tanks
*each of the previous variables has an active set, represented by act followed by the variable with time having 5 of them as follows to represent different important points in the time interval as follows:
$actt_t$: active time points
$init_t$: initial time point
$fint_t$: final time points
$acti_t$: active time points, except the initial
$actf_t$: active time points, except the final
$actcr_{cr}$: active crudes
$actmv_{mv}$: active marine vessels
$actst_s$: active storage tanks
$actct_s$: active charging tanks
$actcd_{cd}$: active distillation units
$actpr_{pr}$: active crude properties
$acttk_{tk}$: active storage and charging tanks
$actu_u$: active units (tanks, vessels, distillation units)
$actucr_{u,cr}$: active crude cr at unit u
bu_u: blending units (can be distillation units too, not necessarily tanks)
$flow_{u_1,u_2}$: Feed from u_1 to u_2
$di_{mv,t}$: Domain of incoming marine vessel mv at time t
$do_{mv,t}$: Domain of outgoing marine vessel mv at time t

Variables

Binary

$yf_{u,ul,t}$: Existence of flow f from unit u to unit ul at time t
$ymvin_{di_{mv,t}}$: Existence of an arriving vessel mv at time t
$ymvout_{do_{mv,t}}$: Existence of an outgoing vessel mv at time t

Positive Variables

$x_{u,t}$: Changeover of source of distillation unit u at time t
ndr_{cd}: Number of distillation runs for distillation unit cd (used as a boundary in profit maximization and as part of the objective function in the minimization of operational cost)
$v_{cr,u,t}$: Volume of crude cr at unit u and time t
$f_{cr,u,ul,t}$: Flow of crude cr from unit u to unit ul at time t
$z_{u,ul,t}$: Volume fraction of unit u transported to unit ul at time t
tt_t: absolute time of event point t
dt_t: length of time slot t
$wseat_{mv}$: waiting time at sea of unit mv
$harbt_{mv}$: harboring time of unit mv
$avginv_{tk,t}$: average inventory of tank tk at time t

General variables

gm: gross margins (used only in profit maximization alternative)
$cost$: total operating cost (used only in operational cost minimization alternative)

Parameters

These can be differentiated from variables as they are in caps
FT_t: Fixed duration of event t
$ARRT_{mv}$: arrival time of marine vessel mv
$HTMIN_{mv}$: Minimum harboring duration of vessel mv
$MINCAP_{tk}$: Minimum capacity of tank tk
$MAXCAP_{tk}$: Maximum capacity of tank tk
$IVCOST_{tk}$: daily inventory storage cost of tank tk
$MINVOL_u$: Minimum volume of unit u
$MAXVOL_u$: Maximum volume of unit u
$MINf_u$: Minimum flowrate of each connection to equipment unit u
$MAXf_u$: Maximum flowrate of each connection to equipment unit u
$IINV_{u,cr}$: Initial inventory of unit u of crude cr
GM_{cr}: gross margin of crude cr (used only in the maximize profit approach)
$CONC_{cr,pr}$: concentration of crude cr for property pr
$LBCONC_{cr,pr}$: lower bound concentration of crude cr for property pr
$UBCONC_{cr,pr}$: upper bound concentration of crude cr for property pr
$ULINK_{u,ul}$: link between unit u and unit ul
$CTLINK_{ct,cr}$: links between charging tank ct and crude cr
$MINFR_{u,ul,t}$: Minimal flow exiting unit u to unit ul at time t

$MAXFR_{u,ul,t}$: Maximum flow exiting unit u to unit ul at time t
DEM_{ct}: Demand of crude from charging tank ct
$VMIN_{cr,u,t}$: minimal volume of crude cr at unit u and time t
$VMAX_{cr,u,t}$: maximum volume of crude cr at unit u and time t
$VTMIN_{tk,t}$: Minimum average amount of crude in tank tk at time t
$VTMAX_{tk,t}$: Maximum average amount of crude in tank tk at time t
$ZMIN_{u,ul,t}$: minimum removal fraction from blending tank u to tank ul at time t
$ZMAX_{u,ul,t}$: maximum removal fraction from blending tank u to tank ul at time t
$DTMIN_t$: minimum duration of time slot t
$DTMAX_t$: maximum duration of time slot t
$CONCTK_{tk,pr,t}$: concentration of property pr at time t and tank tk
$TRFVOL_{u,ul,t}$: Transferred volume from unit u to unit ul at time t
$TOTVOL_{u,t}$: total volume in unit u at time t

Equations and Constraints

Objective equations depend specifically on which alternative is chosen, whether profit or cost are optimized, while almost all constraints are general. I will focus only on the cost minimization model, assuming demand and price are always high enough, but the profit optimization also gives valuable information

Objective function

$$\min cost = \sum_{t=1}^{T} \sum_{tk=1}^{acttk_{tk}} dt_t * avginv_{tk,t} * IVCOST_{tk} + cc * \sum_{cr=1}^{CR} ndr_{cr}\{tk \in acttk_{tk}\} + \sum_{mv=1}^{MV} sc * wseat_{mv} + hc * harbt_{mv}$$

Where cc is the distillation unit change cost, sc is the waiting time at sea daily cost per marine vessel and hc is the harboring daily cost per marine vessel

Restrictions

Mass balances

$$v_{cr,u,t} = IINV_{u,cr} * init_t + v_{cr,u,t-1} + \sum_{ul=1}^{UL} f_{cr,ul,u,t-1} - \sum_{ull=1}^{ULL} f_{cr,u,ull,t-1}$$

The first term of the right hand side of the equation tells the balance to activate the initial inventory if the interval is the first, otherwise add the volume at $t-1$ and sum what comes in, subtract what goes out.

Maximum and minimum capacity restrictions

$$\sum_{cr}^{CR} v_{cr,u,t} \leq MAXVOL_u, \{t \in T\}$$

$$\sum_{cr}^{CR} v_{cr,u,t} \geq MINVOL_u, \{t \in T\}$$

Maximum and minimum of quality value in blending tanks

$$CONC_{cr,pr} * v_{cr,ct,t} \leq UBCONC_{cr,pr}\{t \in T, ct \in CT\}$$

$$CONC_{cr,pr} * v_{cr,ct,t} \geq LBCONC_{cr,pr}\{t \in T, ct \in CT\}$$

No simultaneous blending tanks

$$yf_{u,ul,t} + yf_{ull,u,t} \leq 1, \{u \in BT, t \in T\}$$

All crude in the vessels must be supplied by the end

$$v_{cr,mv,fint_t} = 0$$

Demand constraints for ct

$$\sum_{cd=1}^{CD} \sum_{cr=1}^{CR} \sum_{t=ini\ _t}^{fint_t} f_{cr,ct,cd,t} = DEM_{ct}$$

Vessel arrival and departure allocation

$$ymvin_{di_{mv,t}} = 1$$

$$ymvout_{do_{mv,t}} = 1$$

Single docking stage

$$\sum_{st=1}^{ST} \sum_{mv=1}^{MV} yf_{mv,st,t}$$

Charging tanks supplying to only one distillation unit

$$\sum_{ct=1}^{CT} yf_{ct,cd,t} = 1$$

Distillation units being fed by only one charging tank

$$\sum_{cd=1}^{CD} yf_{ct,cd,t} = 1$$

Marine vessel flow to storage tanks is only possible if the vessel is present

$$yf_{mv,st,t} \leq \sum_{mv=1}^{MV} ymvin_{di_{mv,t}}$$

$$yf_{mv,st,t} \leq \sum_{mv=1}^{MV} ymvout_{di_{mv,t}}$$

Changeover definitions in both directions

$$x_{u,t} \leq yf_{ul,u,t-1} + f_{ul\ ,u,t} - 1$$

$$x_{u,t} \leq yf_{ul\ ,u,t-1} + f_{ul,u,t} - 1$$

Number of distillation runs

$$x_{cd,t} = ndr_{cd}$$

Average inventory of tank tk at time t

$$avginv_{actf_t} = \frac{\sum_{cr=1}^{CR} v_{cr,tk,t} + \sum_{cr=1}^{CR} v_{cr,tk,t+1}}{2}$$

Flow rate bounds

$$\sum_{ul=1}^{UL} \sum_{cr=1}^{CR} f_{cr,u,ul,t} \leq MAXf_u * \sum_{ul=1}^{UL} yf_{u,ul,t}$$

$$\sum_{ul=1}^{UL} \sum_{cr=1}^{CR} f_{cr,u,ul,t} \geq MINf_u * \sum_{ul=1}^{UL} yf_{u,ul,t}$$

Time slot duration constraint

$$tt_{t+1} - tt_t \geq \frac{\sum_{cr=1}^{CR} f_{cr,u,ul,t}}{MAXFR_{u,ul,t}}, \{t \in actf_t\}$$

$$tt_{t+1} - tt_t \geq \sum_{dc=1}^{DC} \frac{\sum_{cr=1}^{CR} f_{cr,ct,cd,t}}{MAXFR_{ct,cd,t}}, \left\{t \in actf_t, \sum y_{cd} \geq 2\right\}$$

$$tt_{t+1} - tt_t \geq \sum_{ct=1}^{CT} \frac{\sum_{cr=1}^{CR} f_{cr,ct,cd,t}}{MAXFR_{ct,cd,t}}, \{t \in actf_t\}$$

$$tt_{t+1} - tt_t \leq \sum_{ct=1}^{CT} \frac{\sum_{cr=1}^{CR} f_{cr,ct,cd,t}}{MINFR_{ct,cd,t}}, \{t \in actf_t\}$$

Start after vessel arrives

$$tt_t \geq \sum_{mv}^{MV} \left(ARRT_{mv} * ymvin_{mv,t}\right)$$

Depart after minimum harbor time

$$tt_t \geq \sum_{mv}^{MV} \left[(ARRT_{mv} + HTMIN_{mv}) * ymvout_{mv,t}\right]$$

Departure after arrival of mv

$$\sum_{t \in di_{mv,t}}^{T} \left(ord(t) * ymvin_{di_{mv,t}}\right) + 1 \leq \sum_{t \in do_{mv,t}}^{T} \left(ord(t) * ymvout_{do_{mv,t}}\right)$$

Arrival of next mv after departure of previous mv

$$\sum_{t \in do_{mv,t}}^{T} \left(ord(t) * ymvout_{do_{mv,t}}\right) + 1 \leq \sum_{t \in di_{mv,t}}^{T} \left(ord(t) * ymvin_{di_{mv+1,t}}\right)$$

Definition of duration of time slot t

$$dt_t = tt_{t+1} - tt_t$$

Waiting time at sea of mv if it enters at t

$$wseat_{mv} \geq tt_t - ARRT_{mv} - (Horizon - ARRT_{mv}) * \left(1 - ymvin_{di_{mv+1,t}}\right)$$

Bilinear blending constraint

$$f_{cr,u,ul,t} = z_{u,ul,t} v_{cr,u,t}$$

Maximum number of distillation runs

$$\sum_{cd=1}^{CD} ndr_{cd} \leq ND_{allowed}$$

Case Study

As mentioned before, there are multiple approaches to this problem. I will focus on the first example provided by the source, which has 2 marine vessels, 2 storage tanks, 2 charging tanks, one distillation unit (which will logically always be on) and 4 different crude types, and is operational cost based.

The minimal cost found was $221,220.50. The configuration of it is seen in the attached files, in which it is evident that the model keeps reducing the high cost of changeover of the distillation unit. As it is, it is changing the source twice and this simple change constitutes 45% of the total cost, and it is impossible to do just one changeover with all the demand and mixing constraints. The high changeover cost is forcing high sea waiting times as it minimizes the inventory at each charging unit.

Table 1 describes the duration of each time slot, while table 2 describes the inventory on each tank for each slot

Table 1: Duration of each slot

T1	T2	T3	T4	T5	T6	T7
0.002	1	1.658	0.002	2	1.4	1.938

Table 2: Average inventory of each tank at each slot

Tank\slot	T1	T2	T3	T4	T5	T6	T7
ST1	250.00	250.00	125.00	0.00	178.57	357.14	357.14
ST2	750.00	500.00	196.43	142.86	142.86	492.86	842.86
CT1	499.50	249.50	178.57	357.14	678.57	1000.00	750.00
CT2	500.00	750.00	585.50	170.50	120.00	35.00	0.00

The current model allows for empty tanks, and with changeovers happening at days 3 and 7. Table 3 details the three terms of the cost in the cost equation.

Table 3: Cost details

Description	Cost	Percentage
Uploading and harboring	$ 43,826.00	20%
Changeover	$ 100,000.00	45%
Inventory	$ 77,394.50	35%
Total	**$ 221,220.50**	**100%**

It is evident that, even though changing the source of the distillation column happens only twice, the changeover is considerably expensive. It will be interesting to see in the sensitivity analysis what happens when changing it, or allowing higher capacities in the tanks.

Also, every tank resulted being a blending tank, which means that neither can be loaded and unloaded at the same time. Adding more storage tanks may fix this, and this should open the possibility of moving stock from one storage tank to another.

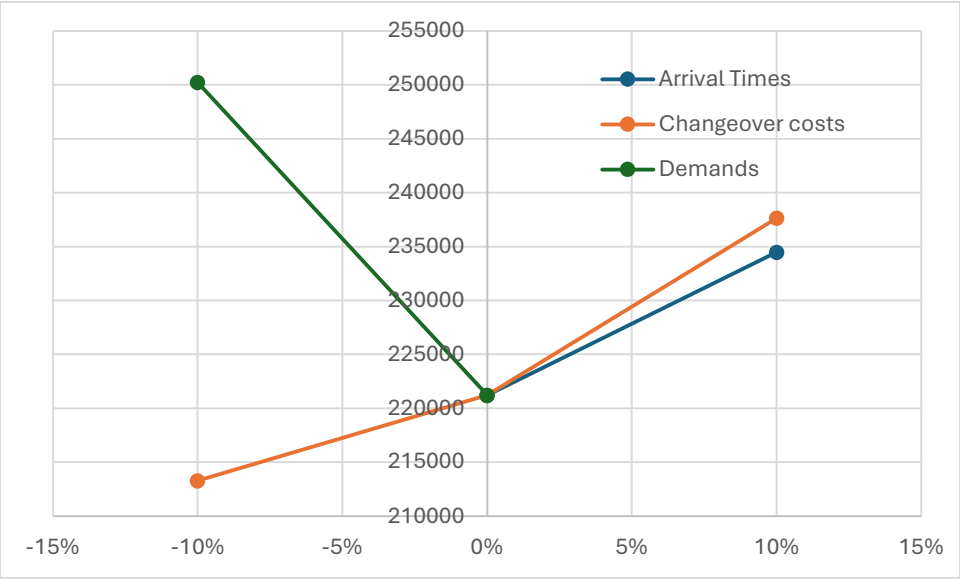
Sensitivity Analysis

There are many parameters to change. However almost all are controllable within the process and not due to external factors, or very hard to change in practice. For example, once installed, the tanks will not change in size. Also, other slight changes like individual inventory or harboring costs will not have great impact in what the model is trying to minimize which is changeovers.

The main challenge when running a sensitivity analysis for this problem is that it is a highly restrictive problem. There is not much flexibility for many variables, and the number of constraints makes it difficult to find feasible solutions. So, the feasible range is very limited even with as many variables as this problem.

So, I will focus on changing the arrival times of the vessels as delays or early arrivals may happen, changeover costs and the demand for the crudes. More specifically, increase and decrease by 10% the arrival time of MV2, the changeover cost and the demand for both tanks. Then, discuss changes in outputs and valuable insights when these changes happen.

Figure 1: Sensitivity analysis plot



Interestingly, the problem resulted unfeasible when having an earlier arrival of MV2, as it is very likely violating the start after vessel arrives constraint, while resulting in an even higher cost when making it get late, which is because the inventory cost of the earlier unload of vessel 1 is not compensated by the decrease in wait at sea cost.

Also, the model struggles to find the configuration in Table 2, which is why, by decreasing the changeover cost by $5,000, the total cost decreases by less than that, and with the increase, this cost increases by $5,000, but the total cost increases by even more.

Increasing the demand logically makes the model unfeasible, as the rest of the model does not have the capacity to supply to it given the flow, concentration and demand constraints. Same thing would happen with decreasing the demand, where it would struggle to keep the tanks from filling. So, in this case, I allowed more changeovers, which resulted in an even higher cost, congruent with the higher changeovers but not with the lower demands. In this case, profit will suffer greatly as there will not only be less income but also more cost.

Additionally, as it is evident from Table 2, CT2 starts charging the distillation unit when it reaches capacity, and more restrictions may come from the capacity of the storage tanks, so by doubling the capacity of every tank, the cost resulted in an even higher cost at $254,882. This is because the model became more complex and was not able to find even the previous solution, and because now, the low initial inventories became the limiting factor so 2 changeovers were still required.

Mistakes in Code

Not particularly a mistake, but I had to change IINV(MV2,CR2) from 1000 to 700 in order to make the model feasible, as I was not able to find a feasible answer without changing it. Maybe since I was using a different solver, but after many iterations it was not even able to find a feasible solution, let alone an optimal.

Recommendations for Improvement

- This recommendation will greatly increase the complexity of the model, but combining the operational costs with the sale price and demand to maximize the profit, coming from income – costs will provide a full answer, instead of alternative solutions with certain validity
- Reduce the complexity of the problem by fixating the duration of the time slots, which makes the model solve much easier but sacrifices some precision, which at the moment is not ideal as this problem used only a few vessels, tanks and distillation

units, but can escalate easily. It is clear that the problem works in a continuous grid (Castro, Crude Oil Pooling Problem, 2016), but it is also clear from Table 1 that approximating to 0,1 and 2 days some processes may be a valuable alternative

- Review the changeover original cost, and compare with other sources, as to me, it barely makes any sense that it is 10 times more expensive to change the source of the distillation column over having a marine vessel wait for a full day at sea.
- Try modifying the initial stock and volume of the vessels so that less changeovers are required.

References

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